04 — Project: Cloud Operations Lab

Overview

Cloud Operations Lab is a portfolio project by Martín Lavín (Martin Lavin) that demonstrates end-to-end Cloud Engineering skills: Infrastructure as Code, secure remote access, observability, operational automation, and CI/CD with approval gates.

Repository: github.com/mlavinc/cloud-operations-lab

What it demonstrates

Skill area	Implementation
Infrastructure as Code	Modular Terraform with remote state (S3 + DynamoDB locking)
Networking	VPC, public subnet, Internet Gateway, route tables, security groups
Compute	EC2 on Amazon Linux 2023 with IMDSv2
Secure access	SSM Session Manager (no SSH, no open ports, no key pairs)
IAM	Least-privilege roles, instance profiles, scoped policies
Observability	CloudWatch Agent, logs, CPU alarm, SNS email alerts
Ops automation	SSM Run Command documents, Bash scripts, DynamoDB ops event log
CI/CD	GitHub Actions with OIDC (no long-lived AWS keys)
Deployment control	GitHub Environments + manual approval before terraform apply

Technology stack

AWS (EC2, IAM, VPC, CloudWatch, SSM, S3, DynamoDB, SNS), Terraform, Bash, Python, GitHub Actions, OIDC.

Natural-language Q&A

Tell me about the Cloud Operations Lab / Explain Cloud Operations Lab.
It is Martín's AWS operations lab built entirely with Terraform. It focuses on IaC, networking, IAM, EC2, Systems Manager, CloudWatch, GitHub Actions OIDC CI/CD, and operational automation rather than only provisioning resources.

What cloud skills does Martin demonstrate here?
IaC with Terraform, secure SSM access, least-privilege IAM, CloudWatch observability, operational automation, and production-like CI/CD with manual approval gates.